MIL-I-15024(SHIPS)
19 September 1952
Used IN LIEU OF
MIL-N-2716
23 August 1951
SUPERSEDING
RE13A516C
15 August 1943

INTERIM MILITARY SPECIFICATION

IDENTIFICATION PLATES, INFORMATION PLATES AND MARKING INFORMATION FOR IDENTIFICATION OF ELECTRICAL, ELECTRONIC AND MECHANICAL EQUIPMENT

1. SCOPE

1.1 Scope. - This specification covers the material requirements for "identification plates" and "information plates" and the marking information for "identification plates" mounted on "major units" and "sets" of Bureau of Ships equipment.

1.2 Classification. - "Identification plates" and "information plates" shall be furnished in the following type plates as specified (see 6.1):

Type A - Etched
Type B - Engraved
Type C - Stamped
Type D - Cast
Type E - Stenciled
Type F - Graphic
Type G - Decalcomanias
Type H - Photographic

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

TT-P-141 - Paint, Varnish, Lacquer, and Related Materials: Methods of Inspection, Sampling and Testing.
TT-C-595 - Colors; (for) Ready-Mixed Paints.

MILITARY

MIL-P-78 - Plastic-Material, Laminated, Thermosetting (for Designation Plates).
MIL-S-854 - Steel, Corrosion-Resisting: Plates, Sheets, Strips, and Structural Shapes.
MIL-N-894 - Nickel-Copper-Alloy; Wrought.
MIL-P-15037 - Plastic-Material, Laminated Thermosetting, Sheets, Glass-Cloth Melamine-Resin.

NAVY DEPARTMENT

General Specifications for Inspection of Material.
47A11 - Aluminum-Alloy (Al-52) (Aluminum-Magnesium-Chromium): Plates, Sheets, and Strips.

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STANDARDS

MILITARY
MIL-STD-12 - Abbreviations for Use on Drawings.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Definitions. -

3.1.1 "Identification plates". - The words "identification plates" wherever used herein shall refer to the identification of "major units" or "sets" of Bureau of Ships equipment.

3.1.2 "Information plates". - The words "information plates" wherever used herein shall refer to instruction plates, wiring diagrams, graphic charts and other designation plates.

3.1.3 "Major unit". - A major unit shall be an electrical, mechanical or hydraulic grouping of parts, subassemblies, assemblies or accessories fixed or joined together into one item.

Examples:

Radar transmitters	Laundry washers
Reduction gears	Dish washers
Internal combustion engines	Garbage grinders
Gyro compass	Printing and photolitho equipment
Loudspeakers	Flatwork ironers
Sonar receivers	Food mixers
Motor generators	Laundry extractors

3.1.4 "Set". - A set shall be a grouping of two or more major units which are electrically, mechanically or hydraulically connected or associated together to perform their intended function.

Examples:

Public address set
Radar set
Diesel generator set

3.2 Materials. -

3.2.1 General. - Identification plates and information plates shall be made of a material that shall withstand the same environmental conditions as the item to which the plates are to be attached. The individual equipment specification shall specify what material is to be used for each application. The general guide for the selecting of material for plates shall be (in order of preference):

(a) Nickel copper alloy (monel) (Specification MIL-N-894).
(b) Brass (commercial).
(c) Corrosion resisting steel (class 9, condition a, finish 1, Specification MIL-S-854).
(d) Aluminum alloy (Specification 47A11).
(e) Plastic (for application not directly exposed to weather)(Specification MIL-P-78 or MIL-P-15037).
(f) Decalcomanias, rubber stamping, stencilling, etc. (for non-metal applications).

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3.2.1.1 Where the scarcity of critical materials prevent the use of nickel-copper alloy, other materials in the order listed shall be used. Brass plates shall not be mounted directly on aluminum. Edges of plastic plates shall be beveled. The surfaces of plastic plates shall have a gloss factor of 25 ± 10 units (Spec. TT-P-141).

3.2.1.2 Color of laminations for plastic plates. - Except for type GMG laminates, the color values of the various lamination for types B, E, and F plastic plates shall conform to Specification MIL-P-78. Colors of cores and cover sheets shall be as specified (see 6.1).

3.2.1.3 Background. - When specified (see 6.1) the background of the plate shall be lusterless orange 3205, conforming to Specification TT-C-595.

3.3 Dimensions for plates. -

3.3.1 Identification plates. - Identification plate dimensions shall depend upon the size of the major unit or set and shall be selected from the standard sizes listed in table I. Adequate space shall be provided to accommodate the information arrangement necessary to suit the format (see 3.7). Standard dimensions for identification plates shall be indicated by size numbers as specified (see 6.1).

Table I - Standard dimensions, identification plates.

Size-number	Length (in.)	Width (in.)	Diameter of holes (in.)	Hole center to edge (in.)	Number of holes
1	2	3/4	1/8	3/16	2
2	2	1	1/8	3/16	2
3	2	2	1/8	3/16	4
4	3	1	1/8	3/16	2
5	3	2	1/8	3/16	4
6	3	3	1/8	3/16	4
7	4	1-1/2	1/8	3/16	4
8	4	2	1/8	3/16	4
9	4	3	1/8	3/16	4
10	4	4	1/8	3/16	4
11	5	2	5/32	3/16	4
12	5	3	5/32	3/16	4
13	5	4	5/32	3/16	4
14	5	5	5/32	3/16	4
15	6	2	5/32	3/16	4
16	6	3	5/32	3/16	4
17	6	4	5/32	3/16	4
18	6	5	5/32	3/16	4
19	6	6	5/32	3/16	4
20	7	2	5/32	3/16	4
21	7	3	5/32	3/16	4
22	7	4	5/32	3/16	4
23	7	5	5/32	3/16	4
24	7	6	5/32	3/16	4
25	7	7	5/32	3/16	4

3.3.2 Information plate dimensions shall be submitted by the contractor to the bureau or agency concerned for approval.

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3.4 Marking information. -

3.4.1 Identification plates. - Marking information to appear on identification plates shall be in accordance with the following subparagraphs outlined and shall be submitted to the bureau or agency concerned for approval. Information which does not pertain to the major unit or set shall be omitted. When the size of the major unit or set restricts the inclusion of the information listed below, only the most essential information shall be included as approved by the bureau or agency concerned.

3.4.1.1 Major unit name and Government type designation. - The major unit name shall be the approved Munitions Board Catalog Agency (MBCA) item name. In the absence of such an approved name, the name in official use may be used. Where a Government type designation includes a name which does not coincide with the MBCA item name, the Government type designation shall prevail. Where space precludes use of the entire major unit name, or set name words, (except the principle noun) may be abbreviated in accordance with Standard MIL-STD-12.

3.4.1.2 Stock number. - The stock number shall be as specified (see 6.1). The source of the stock number shall be indicated. The use of a stock number on an identification plate shall be at the discretion and as specified by the bureau of agency concerned.

3.4.1.3 Serial number. - The serial number shall be entered when required (see 6.1). If a Government serial number is not assigned, the contractor's serial number may be entered upon the approval of the bureau or agency concerned or prime contractors.

3.4.1.4 Manufacturer's or prime contractor's data. - The manufacturer's name, part and drawing or model number shall be entered as applicable when required (see 6.1). This information shall refer to the actual manufacturer or prime contractor. The location of manufacture shall not appear on plates.

3.4.1.5 Contract number. - The procuring activity's contract or purchase order number shall be entered as the contract number. When an order shows both a contract number and a purchase order number, the purchase order number shall be used.

3.4.1.6 U.S. property. - The words "U.S. PROPERTY" shall be inserted at the bottom of the plate.

3.4.1.7 Operating characteristics. - Under this heading shall be entered the information necessary for the safe handling, operation and maintenance of the major unit or set. An additional space or an additional block may be provided as shown on figure 1 for inclusion of this information. The typical operating characteristics for electrical, electronic and mechanical major units and sets shall be as specified in figures 2, 3, and 4 and shall be subject to the approval of the bureau or agency concerned.

3.4.2 Information plates. - Marking information to appear on information plates shall be submitted by the contractor to the bureau or agency concerned for approval.

3.4.3 Marking information, together with views of the complete plate, shall be shown on the working drawings of the equipment concerned.

3.5 Type of marking for plates. - Letters shall be of Gothic capitals, and numerals and other characters shall be of similar appearance. Letters, numerals, and other characters shall be of such size as to be clearly legible. The principal words or group of words and the Government type identification (name of the major unit or set to which identification plate and information plate is to be attached) shall be emphasized by the use of larger letters.

3.5.1 Methods of marking. - The marking shall be made by a method which shall produce permanent and durable markings on identification plates and information plates to withstand the environmental conditions of the major units or sets to which the plates are to be attached. Serial numbers and other designations which vary on each identification plate and information plate may be etched, impression-stamped or punched into the area provided.

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3.5.2 Temporary markings. - The use of decalcomanias, rubber stampings, stencilings or other markings of information of a temporary nature will not be allowed without the specific approval of the bureau or agency concerned, in which case the markings shall be covered with a coat of clear lacquer. Unless otherwise specified in the contract or order, decalcomanias, rubber stampings, stencilings or other markings of information of a temporary nature are not required to withstand the same environmental conditions as the major unit or set to which they are to be attached.

3.5.3 Legibility. - Legibility of the letters, numerals and other characters shall be as necessary for easy readability.

3.5.4 Filling of markings. -

3.5.4.1 All engraved, stamped, or etched letters, numerals and other characters on metal plates shall be well filled with a hard paint, enamel, or lacquer of a contrasting color, as specified (see 6.1) and the face of the plates shall be covered with a moisture-resistant varnish.

3.5.4.2 Plastic plates engraved through one lamination so as to show a lamination of a contrasting color shall not have the letters, numerals and other characters filled.

3.6 Mounting and location. - Identification plates and information plates shall be mounted in a conspicuous place generally on the front panel of the major unit or set. Plates shall be mounted on major units or sets either externally or internally in locations easily accessible to for normal operation and in the event of an emergency. The mounting and location of plates shall be shown on the mechanical assembly drawing of the equipment. In the event the size or nature of the major unit or set precludes the mounting of an identification plate or information plate, information may be marked directly on the major unit or set by means of decalcomanias, rubber stamping, stenciling or other means upon the specific approval of the bureau or agency concerned. The plates shall not be positioned so as to interfere with controls or obscure other required information. Plates shall not be mounted by means of rivets, self tapping screws or welding. Unless otherwise specified in the contract or order, provision shall be made for securely fastening plates with machine screws or hardened steel drive screws. Plates of nickel-copper alloy material in locations exposed to the weather shall be mounted by nickel-copper alloy machine screws. Plates to be mounted on refrigerator major units or sets shall be soft soldered to tank or receiver.

3.7 Format for plates. -

3.7.1 Identification plates. - For guidance purposes only, the arrangement of the marking information for electrical, electronic and mechanical major units or sets shall be as suggested in figures 2, 3, and 4 and shall be submitted by the contractor to the bureau or agency concerned for approval. The format may be varied to suit the major unit or set to which the identification plate is to be attached.

3.7.2 Information plates. - The arrangement of the marking information shall be submitted by the contractor to the bureau or agency concerned for approval.

3.8 Type A (etched). -

3.8.1 Material. - Type A plates shall be of one of the following materials:

Corrosion-resisting steel.
Nickel-copper alloy.
Aluminum alloy.
Brass.

3.8.2 Marking. - Letters, numerals, and other characters shall be etched. Additional information if necessary may be stamped on the plate.

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3.8.3 Dimensions. - Etched letters, numerals and other characters shall not be less than 0.003 inch deep. Stampings of additional information shall not be less than 0.003 inch deep. The thickness of the A plates shall be as follows:

Corrosion resisting steel - 0.03 inch
Nickel copper alloy - 0.03 inch
Aluminum alloy - 0.05 inch
Brass - 0.05 inch

3.9 Type B (engraved). -

3.9.1 Materials. - Type B plates shall be of one of the materials specified in 3.8.1 or of plastic material conforming to type NDP of Specification MIL-P-78 or Specification MIL-P-15037. Other types of plastic material for a specific purpose may be used upon the approval of the bureau or agency concerned.

3.9.2 Marking. - Letters, numerals and other characters shall be engraved. Additional information if necessary may be stamped on the plate.

3.9.3 Dimensions. - The depth of engraving and thickness shall be as specified hereinafter. Stampings of additional information shall not be less than 0.003 inch deep.

Plate material	Minimum depth of engraving (inch)	Thickness (inch)
Plastic	0.003	0.0625
Brass or aluminum alloy	.003	.05
Nickel-copper alloy	.003	.03
Corrosion-resisting steel	.003	.03

3.10 Type C (stamped). -

3.10.1 Materials. - Type C plates shall be of one of the following materials:

Brass.
Nickel-copper alloy.
Corrosion-resisting steel.

3.10.2 Marking. - Letters, numerals and other characters shall be stamped on the plate. Embossing or relief stamping is not acceptable.

3.10.3 Dimensions. - Stamping shall not be less than 0.003 inch deep. Thickness of type C plates used shall be in accordance with 3.8.3.

3.11 Type D (cast). -

3.11.1 Materials. - Type D plates shall be of cast brass or bronze of commercial quality.

3.11.2 Marking. - Letters, numerals and other characters shall be raised above the body of the plate and shall be polished, the balance of the plate to have a roughened or stippled finish. Additional information if necessary may be stamped on the plate.

3.11.3 Dimensions. - Letters, numerals and other characters shall be raised to 0.03 inch. Thickness of type D plates shall be as specified (see 6.1).

3.12 Type E (stenciled). -

3.12.1 Material. - Type E plates shall be of plastic material conforming to type HSP of Specification MIL-P-78 or Specification MIL-P-15037. Other types of plastic material for a specific purpose may be used upon the approval of the bureau or agency concerned.

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3.12.2 Marking. - Letters, numerals and other characters shall be stenciled on the plate. Marking shall be applied by lithographing, by the silk screen method (method of printing through a stencil that is bonded to a silk screen) or a similar process.

3.12.3 Dimensions. - Thickness of type E plates shall be 0.0625 inch.

3.13 Type F (graphic). -

3.13.1 Material. - Type F plates shall be of plastic material, type GCP-H (Specification MIL-P-78). Other types of plastic material for a specific purpose may be used upon the approval of the bureau or agency concerned.

3.13.2 Marking. - The printed information shall be included between the laminations and shall be clearly legible through a transparent outer layer.

3.13.3 Dimensions. - Thickness of type F plates shall be 0.0625 inch.

3.14 Type G (decalcomanias). -

3.14.1 General. - Decalcomanias shall be mounted on major units normally considered expendable and which do not require permanent marking information to withstand the environmental conditions of the major unit or set. Decalcomanias in general unless otherwise specified in the contract or order. shall be used to indicate the information necessary for delivery, handling, preliminary and/or final installation and/or operation of the major unit or set. Decalcomanias may also be used upon specific approval of the bureau or agency concerned on major units or sets subject to radioactive contamination.

3.14.2 Materials. - The plates shall be made of materials supplied by manufacturer's of decalcomanias approved by the bureau or agency concerned. Decalcomania materials shall depend upon the type of material, surface finish and exposure of the major unit or set to which the decalcomania is to be affixed.

3.14.3 Marking. - Letters, numerals and other characters shall be of the open letter design or solid letter design. The open letter design shall have the background omitted and the solid letter design shall have the letters, numerals and other characters integral with the background. Additional markings may be rubber stamped or lettered directly on the decalcomania before mounting.

3.14.4 Dimensions. - Thickness of type C plates shall depend upon the material and the expected life endurance of the decalcomania necessary for the delivery, handling, preliminary or final installation, and operation of the major unit or set to which the decalcomania is to be affixed.

3.14.5 Mounting. - Decalcomanias shall be designed for mounting on metallic, phenolic or organic surface finishes. Mounting of decalcomanias shall be in accordance with standard application procedures specified by the manufacturer of decalcomanias and approved by the bureau or agency concerned. Properly applied decalcomanias shall be firmly adhered and free of bubbles or loose edges. Should a transfer be damaged beyond usefulness after its application, a second transfer may be applied directly over the first transfer.

3.15 Type H (photographic). -

3.15.1 Materials. - Type H plates shall be of one of the following materials depending upon the photographic process used:

Corrosion-resisting steel
Aluminum

3.15.2 Marking. - Letters, numerals and other characters shall be integrated into the plate by a photographic process approved by the bureau or agency concerned and as specified (see 6.1). Stamping of additional marking information will not be permitted on type H plates.

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3.15.3 Dimensions. - Thickness of type H plates shall be as follows:

Plate material	Thickness (inch)
Corrosion-resisting steel	0.0156
Aluminum	0.02

3.16. Workmanship. - Identification plates and information plates shall be furnished with smooth edges and shall be free from sharp corners. Unless otherwise specified in the contract or order, identification plates and information plates shall be suitable for mounting on finished major units or sets.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Inspection and tests shall be conducted to determine that the material employed is in accordance with the requirements of the applicable material specification. Inspection and tests shall also be conducted to determine that the letters, numerals and other characters are of the specified height on cast identification plates and information plates or are stamped, etched, or engraved to the depth required. In determining the depth of stamping, etching, or engraving, care should be exercised in removing the filling material to insure that all material is removed. Surface inspection shall also be conducted to insure that the identification plates and information plates furnished comply with the requirements of the contract or order regarding dimensions, thickness, format, life endurance and finish. No chemical analysis will be required for brass or bronze plates. Plates necessary for proper inspection shall be furnished in addition to those required by the contract or order and without charge to the Government. One production run sample of each type of identification plate or information plate being supplied on the contract or order shall be furnished for inspection. At the request of the bureau or agency concerned the contractor shall furnish a photographic print of each type identification plate or information plate being supplied in lieu of a production run sample.

5. PREPARATION FOR DELIVERY

5.1 There are no packaging, packing, and marking requirements applicable to this specification.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether an identification or information plate is required and the type (see 1.2).
(c) Material required (see 3.2.1, 3.8.1, 3.9.1, 3.10.1, 3.11.1, 3.12.1, 3.13.1, 3.14.1 and 3.15.1.
(d) Color of cores and cover sheets (see 3.2.1.2).
(e) Whether an orange background is required (see 3.2.1.3).
(f) Size number (see 3.3.1).
(g) Stock number (see 3.4.1.2).
(h) Serial number if required (see 3.4.1.3).
(i) Manufacturers or prime contractor's data if required (see 3.4.1.4).
(j) Color of filler for metal plates (see 3.5.4.1).
(k) Photographic process for type H plates (see 3.15.2).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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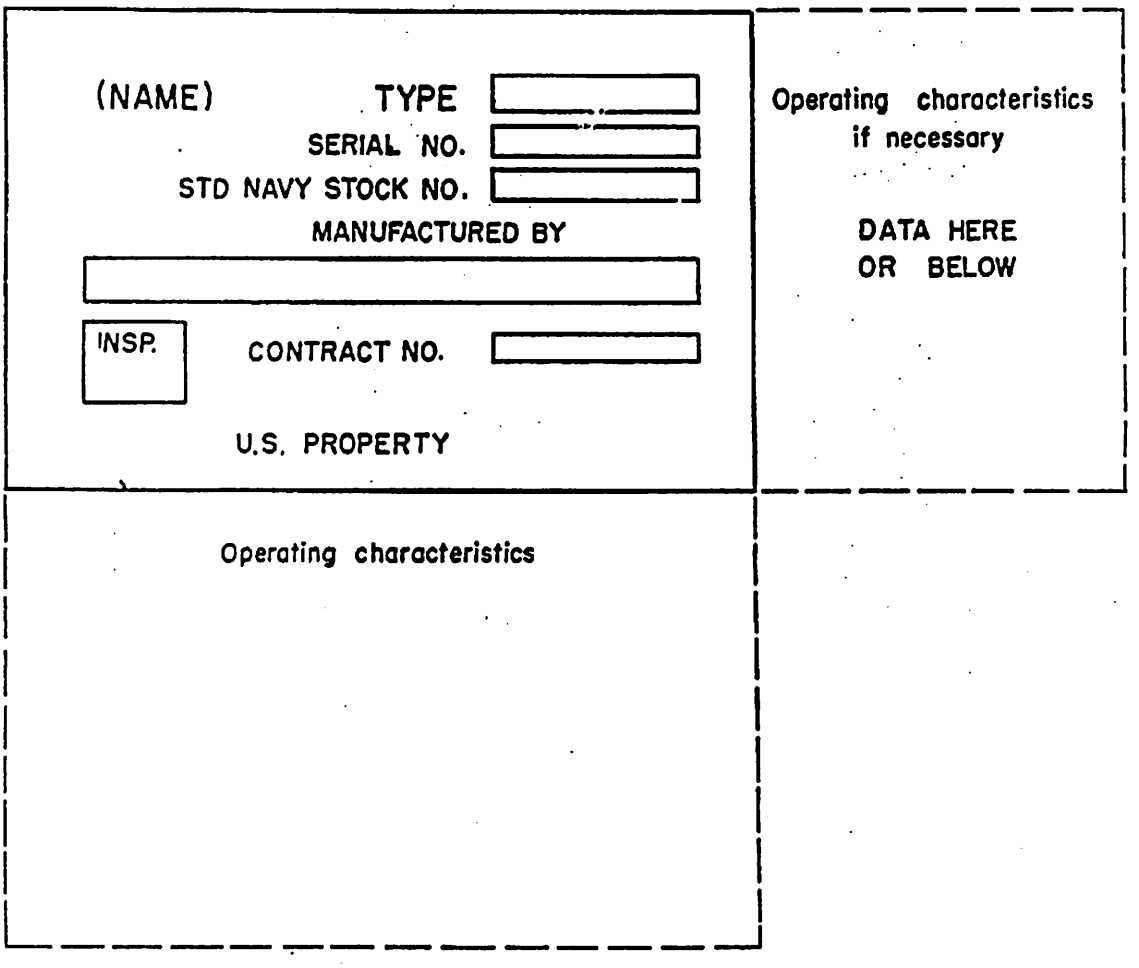


Figure 1.– Identification plates– General use, format showing typical operating characteristics.

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MAJOR UNIT

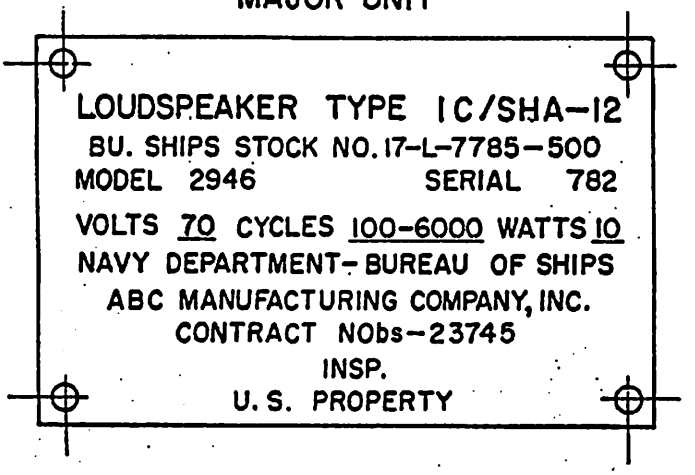


Figure 2 A.—Typical IC/FC equipment.

MAJOR UNIT FOR SET

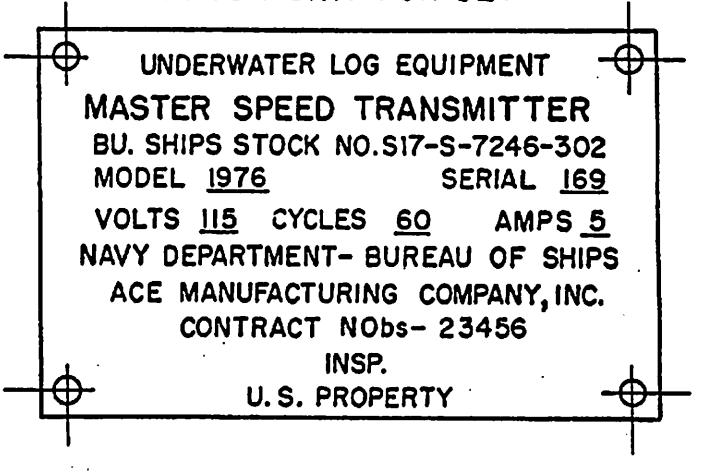


Figure 2 B.—Typical IC/FC equipment.

Figure 2.— Identification plates, electrical, format showing typical operating characteristics.

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Figure 3.—Identification plates-Electronic, format showing typical operating characteristics

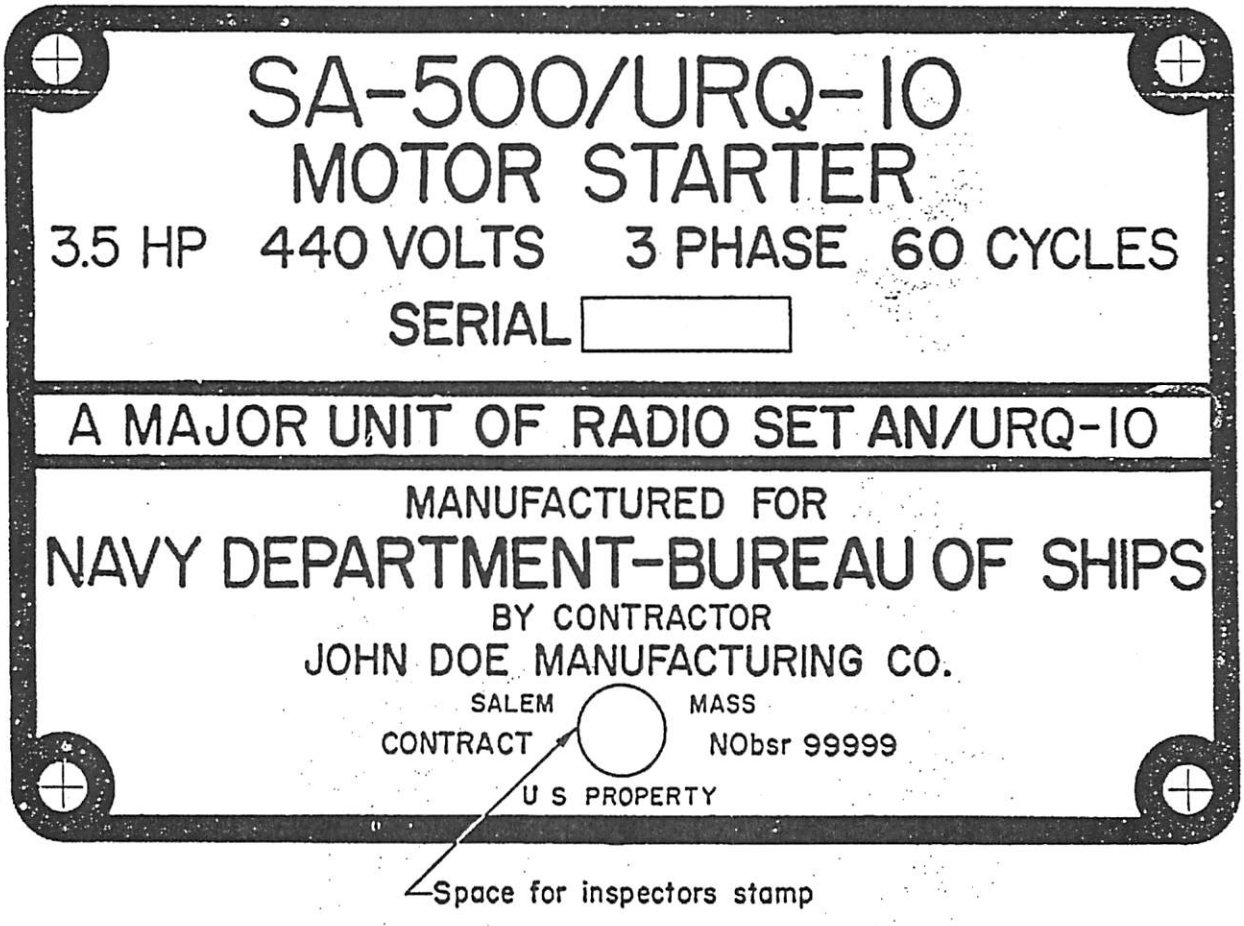
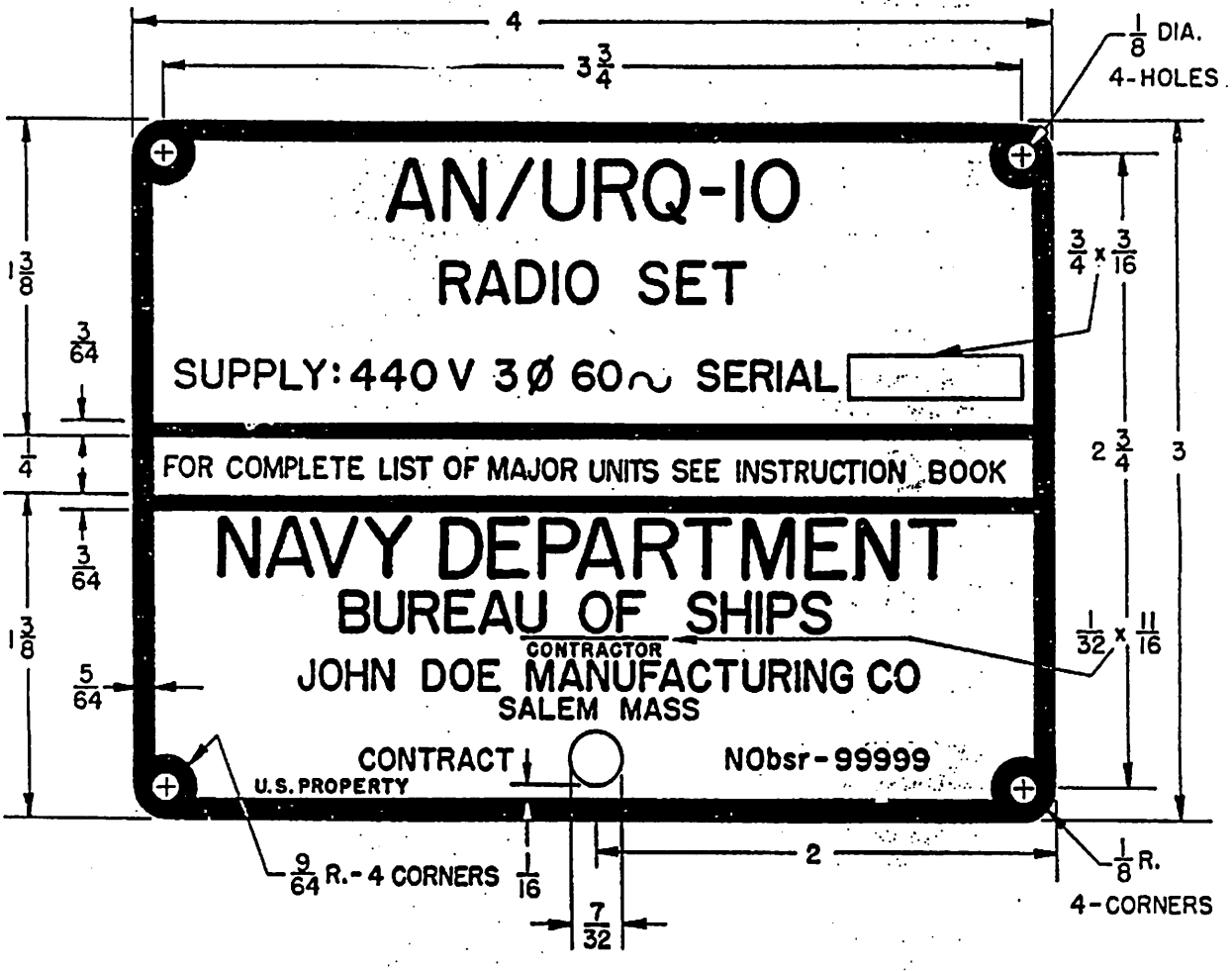


Figure 3A.—Major unit (typical ship, shore and amphibious equipment.)

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NOTE - Dimensions shown are typical to show proportions and spacing.

Figure 3B. - Set

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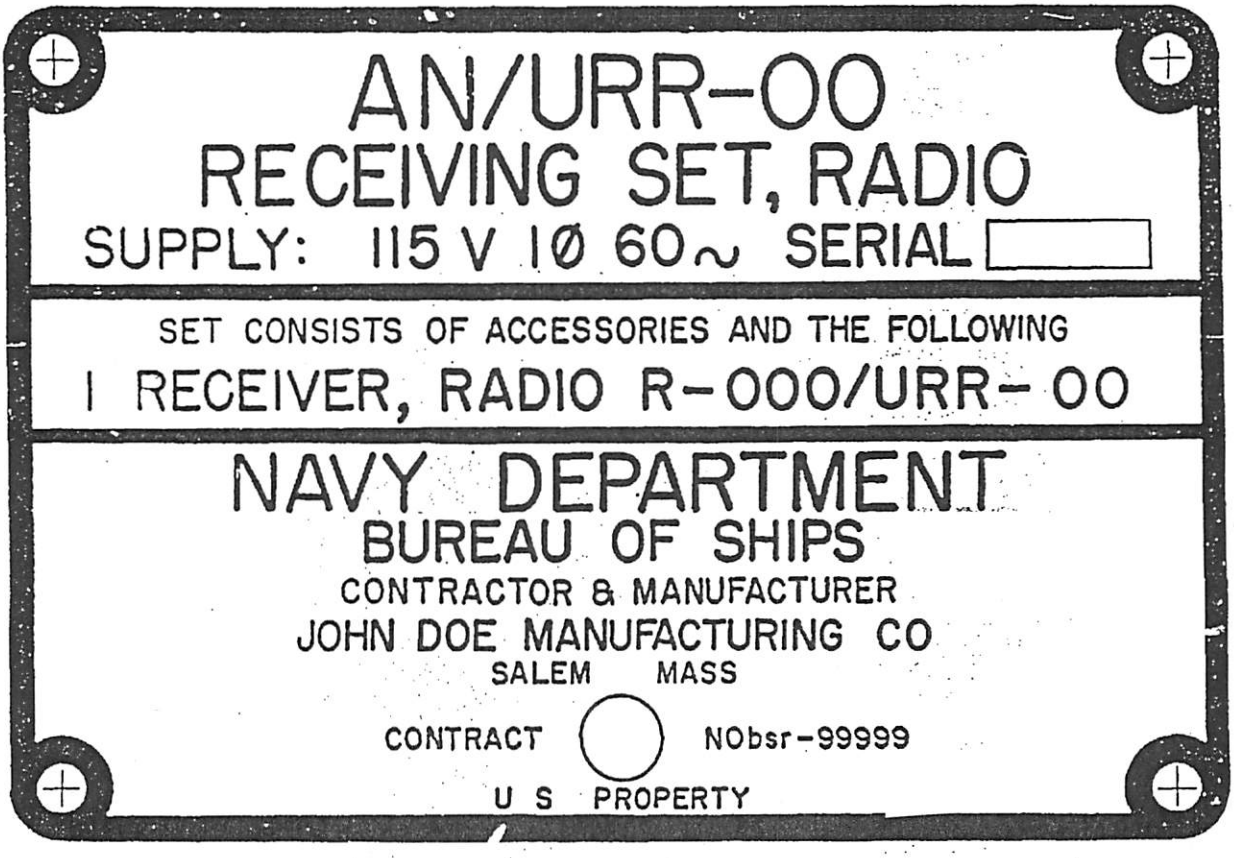


Figure 3 C.- Set and major unit-combination.

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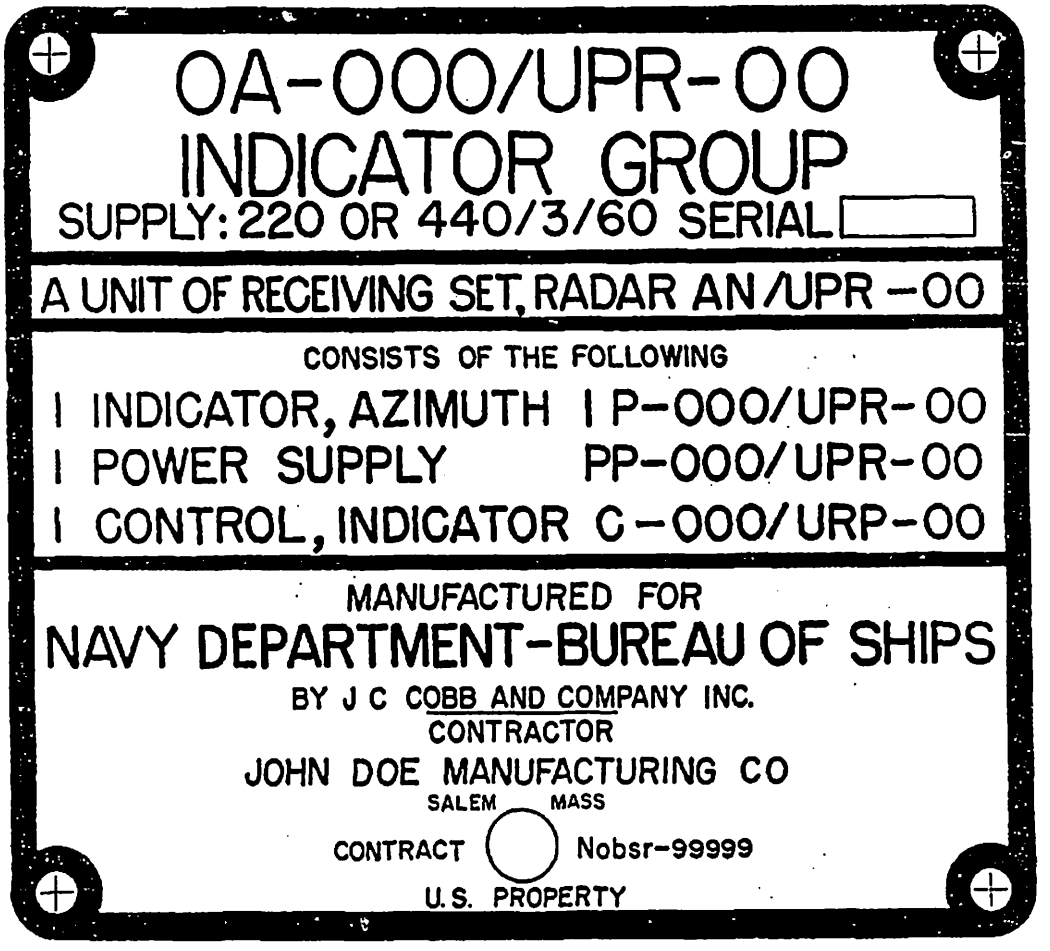


Figure 3D.- Set.

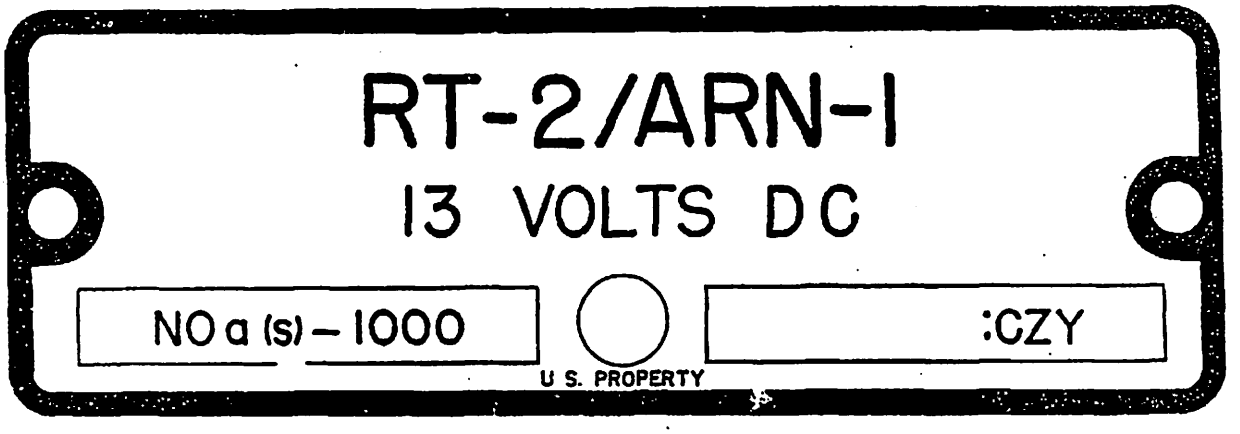


Figure 3E.- Major unit (Airborne Equipment)

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Figure 4.– Identification plates–Mechanical, format showing typical operating characteristics.

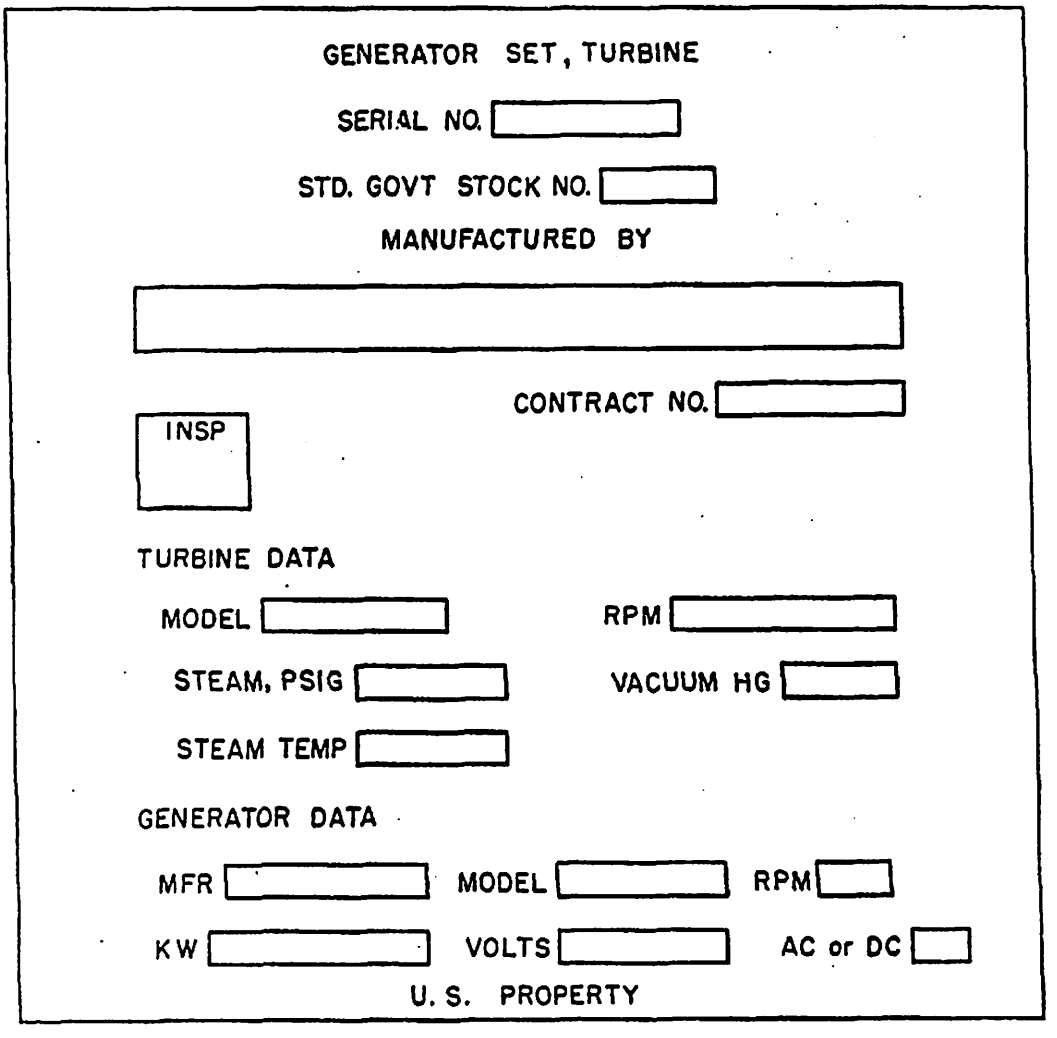


Figure 4 A– Set.

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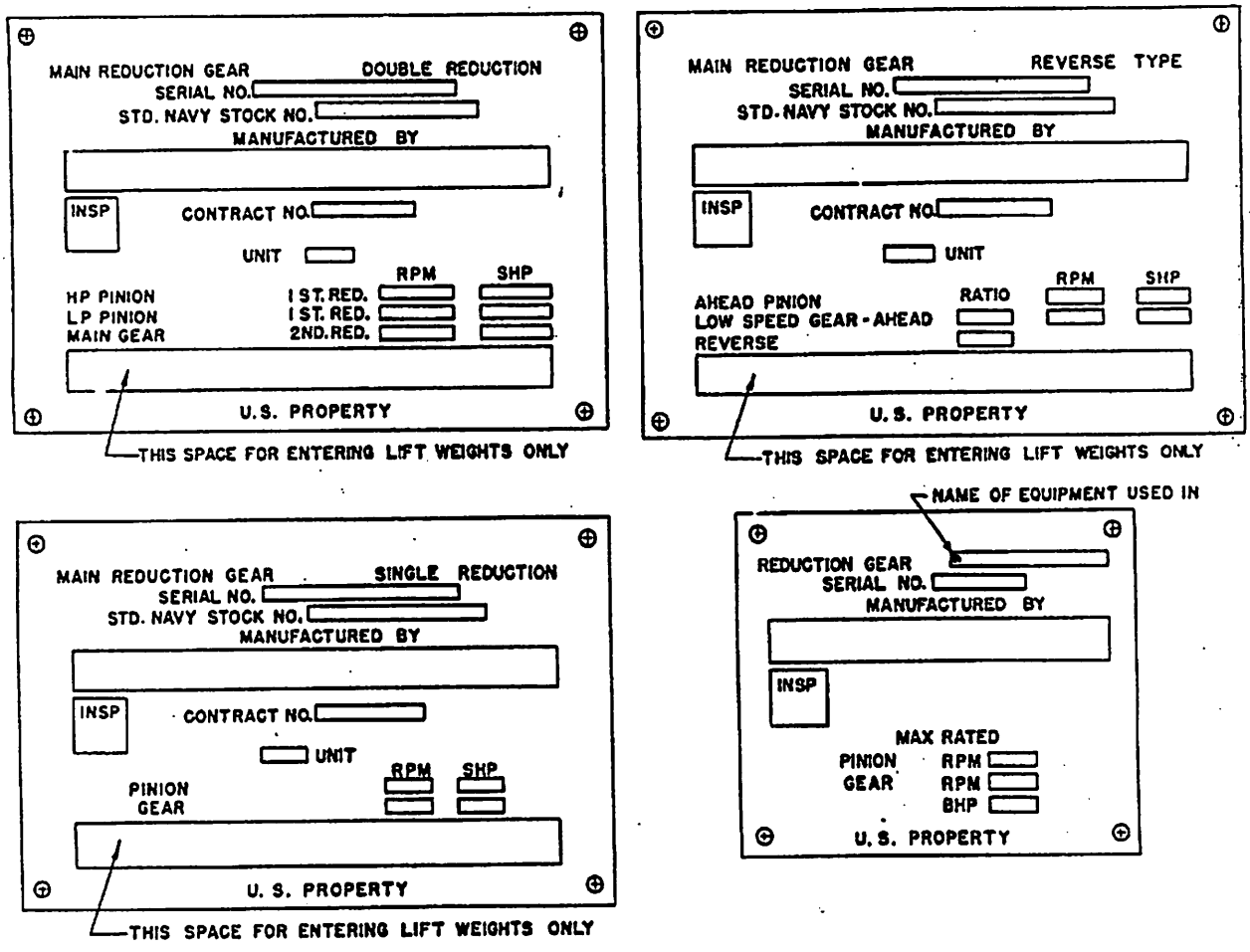
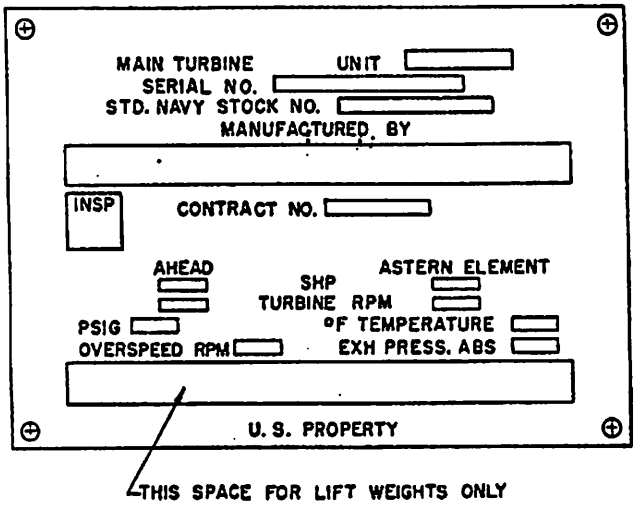


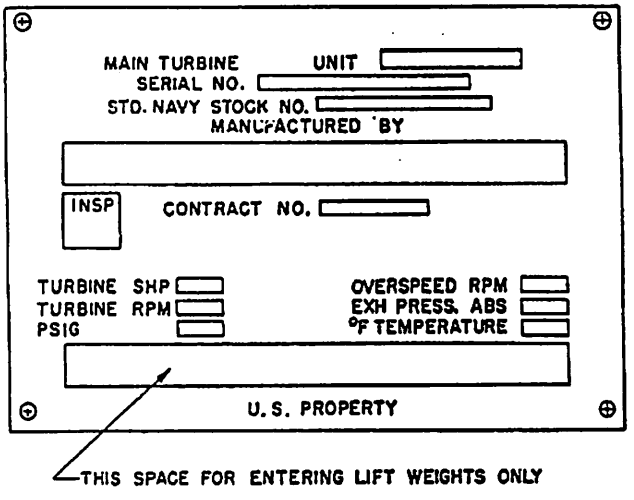
Figure 4B.- Typical reduction gears major unit

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IDENTIFICATION PLATE FOR COMBINED LP AND ASTERN TURBINE



IDENTIFICATION PLATE FOR CRUISING, HP AND ELECTRIC DRIVE TURBINE



NAME OF EQUIPMENT TO BE USED IN

IDENTIFICATION PLATE FOR AUXILIARY TURBINE

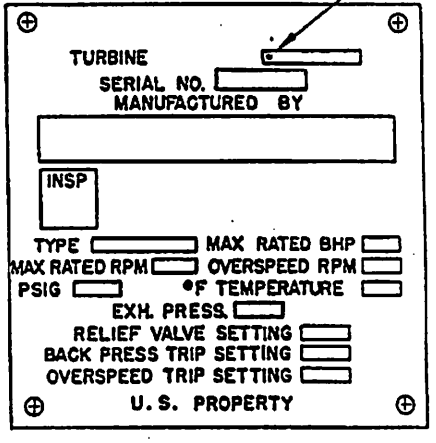
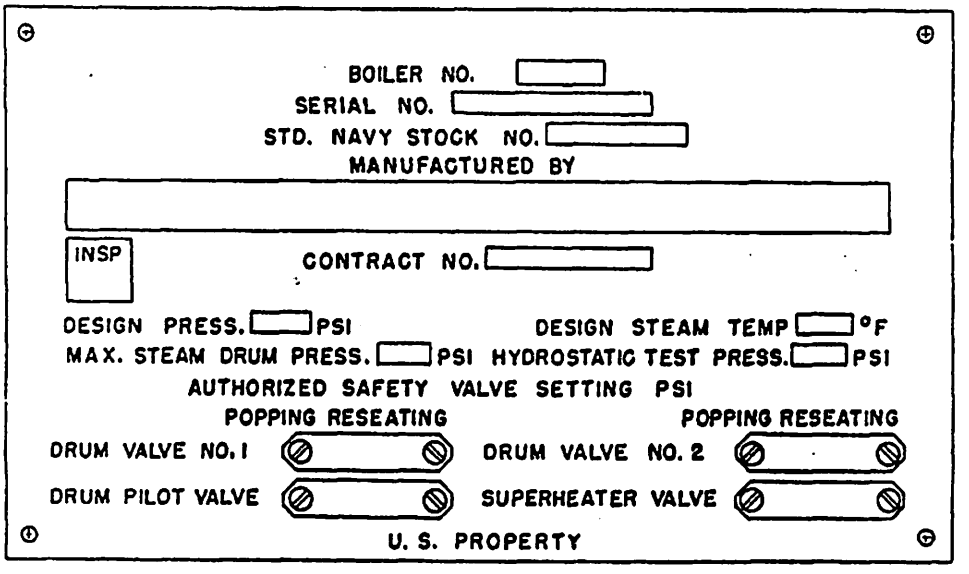


Figure 4 C.– Major unit typical turbines.

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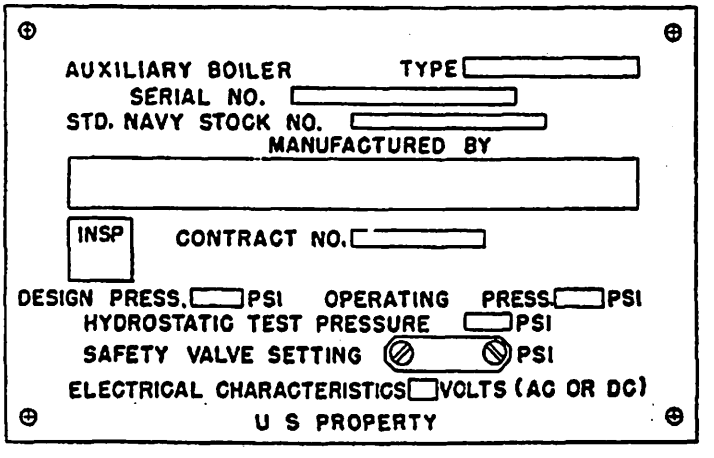
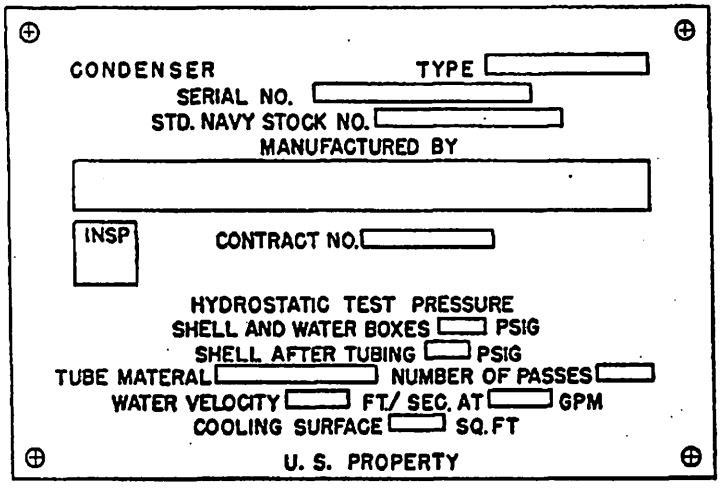
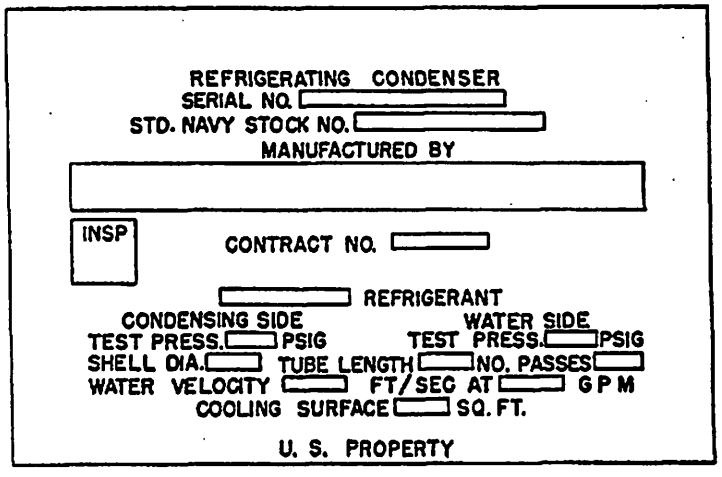


Figure 4D. - Major unit typical boilers

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Note: Identification plates shall be soft soldered to tank or receiver.

Figure 4 E.- Major units typical condensers.

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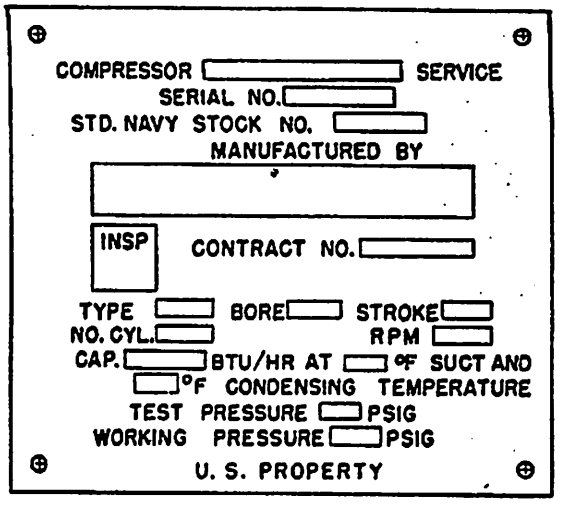
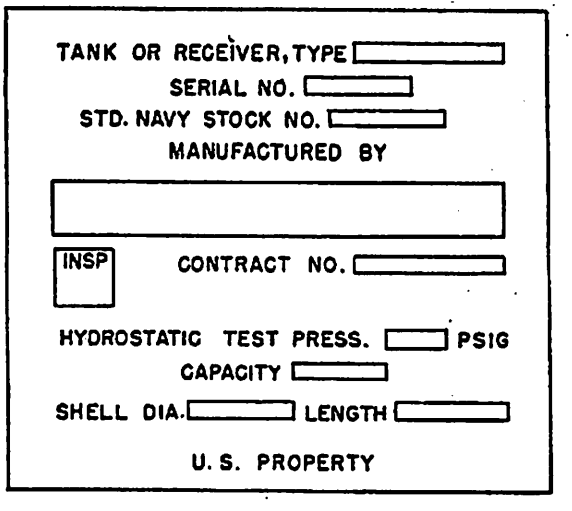


Figure 4F.



Note: Identification plate shall be soft soldered to tank or receiver.

Figure 4G.

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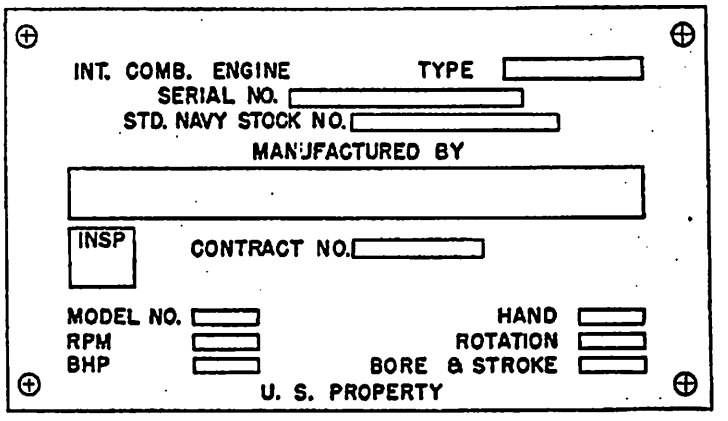


Figure 4H

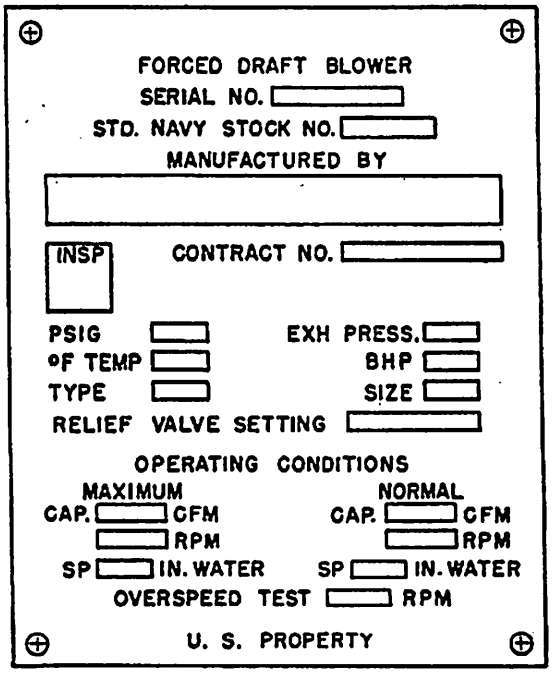


Figure 4 J

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